

SS1 Mathematics Diagnostic: General Core (Part 1)

Target: All SS1 Students (Science, Art, Commercial)
Focus: Junior School Mastery & Senior School Bridge (Questions 1 - 20)

Mathematics - Common Core

No.	Question & Options	Ans	Diagnostic Purpose
1	If $P = \{1, 2, 3, 4, 5\}$ and $Q = \{2, 4, 6, 8\}$, find $P \cap Q$. A) $\{2, 4\}$ B) $\{1, 3, 5\}$ C) $\{1, 2, 4\}$ D) $\{1, 2, 3, 4, 5, 6, 8\}$	A	Set Theory (Common intersection).
2	The sum of any two prime numbers is always even. Which of the following is a counter-example? A) $3+5$ B) $2+3$ C) $7+11$ D) $13+17$	B	Mathematical Logic & Prime Properties.
3	Evaluate $(16)^{\frac{3}{4}}$. A) 4 B) 8 C) 12 D) 16	B	Indices (Fractional powers).
4	Solve for x : $2(x - 3) = \frac{x}{2} + 3$. A) $x = 6$ B) $x = 9$ C) $x = 4$ D) $x = 12$	A	Algebraic Equations (Fractional).
5	Calculate the Simple Interest on N40,000 for 3 years at 5% per annum. A) N6,000 B) N2,000 C) N8,000 D) N1,200	A	Financial Literacy (Essential for all).
6	If $\tan \theta = \frac{3}{4}$, find $\cos \theta$. A) $\frac{4}{5}$ B) $\frac{3}{5}$ C) $\frac{5}{4}$ D) $\frac{1}{2}$	A	Trigonometry foundations (3-4-5 triangle).
7	Find the range of the set of numbers: 12, 18, 10, 25, 15, 30. A) 15 B) 20 C) 30 D) 10	B	Statistics (Dispersion basics).
8	Convert 1101_{two} to base ten. A) 11 B) 13 C) 15 D) 9	B	Number Bases (ICT readiness).
9	Factoring $2x^2 - 8$ completely gives: A) $2(x-2)(x+2)$ B) $(2x-4)(x+2)$ C) $2(x-4)$ D) $2(x-2)^2$	A	Difference of two squares (SS1 readiness).
10	A bag contains 5 red, 3 blue, and 2 green marbles. Find the probability of NOT picking a blue marble. A) 0.3 B) 0.7 C) 0.5 D) 0.8	B	Probability of complements.
11	What is the value of $(0.125)^{\frac{1}{3}}$? A) 0.5 B) 0.25 C) 0.05 D) 0.1	A	Indices & Decimals.
12	Subject of Formula: If $V = \frac{1}{3}\pi r^2 h$, find h in terms of V , π and r . A) $h = \frac{3V}{\pi r^2}$ B) $h = \frac{V}{3\pi r^2}$ C) $h = 3V\pi r^2$ D) $h = \frac{\pi r^2}{3V}$	A	Formula manipulation (Critical for Physics/Accounting).
13	Calculate the mean of the first five even numbers. A) 4 B) 5 C) 6 D) 10	C	Statistics & Sequence property.
14	In a survey of 40 students, 25 like Math and 20 like English. If 5 like neither, how many like both? A) 10 B) 15 C) 5 D) 20	A	Set Theory (Word problem).
15	Find the gradient of the line passing through points $(2, 3)$ and $(5, 9)$. A) 2 B) 3 C) 6 D) -2	A	Coordinate Geometry foundations.
16		A	Surds (Introduction for SS1).

No.	Question & Options	Ans	Diagnostic Purpose
	Simplify $\sqrt{75} - \sqrt{12}$. A) $3\sqrt{3}$ B) $2\sqrt{3}$ C) $\sqrt{63}$ D) $5\sqrt{3}$		
17	If a trader buys an item for N2,500 and sells it for N3,000, what is the percentage profit? A) 20% B) 25% C) 15% D) 30%	A	Arithmetic (Profit/Loss).
18	An interior angle of a regular polygon is 140° . How many sides does it have? A) 8 B) 9 C) 10 D) 12	B	Polygons (Exterior-interior relationship).
19	Evaluate $\log_{10} 1000$. A) 2 B) 3 C) 4 D) 10	B	Logarithms basics (Senior secondary bridge).
20	Diagnostic Insight: Students who fail Q9 and Q12 will struggle in Science (Physics) and Commercial (Accounting) tracks.	-	Pathway placement.